Rafael Velecela

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EDUCATION

University of South Florida | *B.S. in Computer Science* **Expected May 2027**

- Cumulative GPA: 3.75/4.00
- Relevant Coursework: Algorithms, Trustworthy Infrastructure, Software Engineering

SKILLS & INTERESTS

Programming Python, C, COBOL

AI & Data Retrieval-augmented generation, semantic search, document embeddings, Agents, local LLM inference

Tools Ubuntu/WSL2, Git, GitHub, Docker, Docker Compose, FastAPI

Hardware Raspberry Pi

Interests Photography and videography; AI research; motorsport engineering and simulation; integrated hardware-software systems

PROJECTS

RAG Documentation Retrieval | *GitHub* **Ongoing**

- Tech: Python, FastAPI, Qdrant, Docker, Docker Compose.
- Built a FastAPI endpoint that queries Qdrant and returns vector-database collection metadata.
- Defined a two-service Docker Compose stack with locally bound ports and persistent Qdrant storage.

Artemis: Offline AI Voice Assistant | *GitHub* **2026**

- Tech: Python, Raspberry Pi 5, Qwen 2.5, llama.cpp, SAM, MQTT, Mosquitto.
- Built an offline-first assistant with fuzzy intent, topic, and severity routing plus token-budgeted knowledge selection for local inference.
- Integrated authenticated, read-only MQTT sensor data with stale/offline validation and deterministic temperature and humidity answers.
- Automated Mosquitto setup, least-privilege ACL provisioning, retained-message verification, and access-control tests.

InCollege | *COBOL* **2026**

- Implemented local user-search functionality for a COBOL console application.
- Validated teammates' features against expected behavior and identified functional issues.
- Worked across Scrum Master, developer, and tester roles in a five-person Agile team.